

VALIDATION_EVIDENCE

Helix OTA 1.0.0-MVP — Artifact Validation Evidence

Field	Value
Revision	1
Created	2026-06-07
Last modified	2026-06-07
Status	active
Status summary	Physical, reproducible validation of the executable/parseable 1.0.0-MVP artifacts (OpenAPI, SQL migrations, Kubernetes manifests, docker-compose) using real tools on the build host. Per HelixConstitution §7.1/§11.4.6/§11.4.123 — no format/artifact is claimed valid unless a real validator confirmed it; commands + outputs are reproduced here.
Issues	docker engine absent on host → docker-compose validated by YAML parse only (not docker compose config). Kotlin snippets are design references (not compiled — no Android SDK on host); flagged UNVERIFIED for compilation.
Issues summary	OpenAPI, SQL (live Postgres), and k8s (kubeconform) are fully verified.
Fixed	initial validation run
Fixed summary	All parseable/executable artifacts validated; results captured below.
Continuation	Add a CI job running these exact validators (redocly, a Postgres service applying migrations up+down, kubeconform, docker compose config) as a §1 source/artifact gate; compile Kotlin snippets once the

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	AOSP/Android-SDK toolchain is wired.

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§1. OpenAPI 3.1 — redocly lint

Command: `npx --yes @redocly/cli@latest lint api/openapi.yaml`

Result: **valid** — Woohoo! Your API description is valid. 🎉 You have 1 warning. The single warning is info-license (recommended license field absent) — cosmetic, to be added. Structure (python yaml): `openapi: 3.1.0, 12 paths (/auth/login, /auth/refresh, /devices/register, /devices/{deviceId}/status, /artifacts/upload, /artifacts/{artifactId}, /releases, /releases/{releaseId}, /deployments, /deployments/{deploymentId}, /client/update, /client/telemetry), 24 component schemas`.

§2. PostgreSQL migrations — applied to a live server

A live PostgreSQL server was available (`pg_isready → /tmp:5432 - accepting connections`). A scratch database was created; both migrations applied with `psql -v ON_ERROR_STOP=1`:

- `001_initial_schema.up.sql → UP OK`. \dt helix_ota.* listed **12 tables**: `api_keys, artifact_versions, artifacts, audit_logs, deployments, device_deployments, device_group_members, device_groups, devices, releases, telemetry_events, users`.
- `001_initial_schema.down.sql → DOWN OK`. \dt helix_ota.* → *Did not find any relation* (clean teardown).
- Scratch database dropped.

This proves the DDL is syntactically and referentially valid (FKs, CHECKs, indexes all created) and the down-migration is a correct inverse.

§3. Kubernetes manifests — kubeconform

Command: `kubeconform -summary -strict deployment/kubernetes/*.yaml`

Result:

5 resources found in 4 files – Valid: 5, Invalid: 0, Errors: 0, Skipped: 0 (namespace, deployment, service, statefulset).

§4. docker-compose — YAML parse

Docker engine is not installed on the host, so `docker compose config` could not run. The file parses as valid YAML with services: `ota-server`, `postgres`, `minio`, `reverse-proxy`, `dashboard`. Full `docker compose config` validation is deferred to CI where the engine is present.

§5. Tool versions / host

`pandoc`, `mmdc`, `python3`, `psql` (server live), `kubectl`, `kubeconform`, `npx/node22`, `java17` present; `docker`, `yq`, `swagger-cli`, `sqlfluff`, `LaTeX engine`, `drawio`, `plantuml`, `dot` absent (validations needing them are deferred to a CI container, not faked).